CardioTriage 3D

Vessel-level coronary stenosis prediction with interactive 3D risk mapping
Multimodal AI Hackathon 2026 — Track A: Cardiovascular Risk Visualization & Prediction

Team: [TODO: names] | Repo: [TODO: GitHub URL] | Live demo: [TODO: URL] | Video: [TODO: YouTube URL]

1. Overview

Cardiovascular risk tools usually output a single percentage. That does not tell a clinician *which* coronary vessel is implicated. CardioTriage 3D trains four binary classifiers on the Extension of the Z-Alizadeh Sani dataset (overall CAD, LAD, LCX, RCA stenosis) and displays the predicted probabilities as colour on the corresponding coronary arteries of an interactive 3D heart, next to a per-patient SHAP explanation and a what-if sensitivity panel.

Design principles

- **No target leakage:** Cath, LAD, LCX and RCA are removed from the input matrix and an assertion enforces it at training time.
- **Honest validation:** stratified cross-validation, out-of-fold metrics reported for all four targets with confidence intervals.
- **Spatial correctness:** only the model outputs LAD/LCX/RCA are mapped to vessels. Non-spatial features (e.g. Region with RWMA) are never converted into 3D coordinates; they enter the model as ordinary tabular inputs.
- **Extensible:** features, models and anatomical structures are config-driven (Section 5).

System at a glance

Layer	Implementation
Data / ML	Python, pandas, scikit-learn, XGBoost, SHAP (TreeExplainer)
Backend	FastAPI + Uvicorn + Pydantic; endpoints /api/init, /api/predict, /api/parse_report
Frontend	Vanilla ES6 + Three.js (OrbitControls, Raycaster); no build step
3D assets	[TODO: state exact mesh source, license and attribution, e.g. BodyParts3D / CC-BY .glb; or state that geometry is procedural]
Deployment	Render.com ([TODO: Docker? keep-alive/warm-up configured?])

Intended use. Educational and decision-support demonstration only. It is not a medical device, has not been externally validated, and must not replace coronary angiography or clinical judgement.

2. Dataset and Preprocessing

Source. UCI Machine Learning Repository, Extension of Z-Alizadeh Sani (ID 411): 303 patients from a single centre. Features span demographics, examination, ECG, laboratory and echocardiographic findings. CAD is defined as ≥50% diameter stenosis in a major vessel.

Class balance. [TODO: insert counts: CAD positive/negative; LAD, LCX, RCA positives]. Small positive counts for LCX and RCA make estimates for these vessels noisy; this is reflected in the confidence intervals in Section 4.

Preprocessing steps

- Load the Excel file; trim whitespace and fix categorical typos/inconsistent labels ([TODO: list the exact fixes]).
- Encode binary and categorical columns (Y/N, Sex, ECG findings, RWMA region) to numeric codes; [TODO: describe encoding of Region with RWMA and other multi-level fields].
- Missing values: [TODO: state whether any exist and how handled]. Imputation, if any, is fit inside each training fold only.
- Feature count: [TODO: confirm 54 from your code, e.g. len(X.columns)] after dropping targets.

Target-leakage prevention

LAD, LCX, RCA and Cath are label columns for the same angiographic result. Using any of them as an input would let the model read the answer. The training script drops them and asserts the result:

```
TARGET_COLS = ["Cath", "LAD", "LCX", "RCA"]
X = df.drop(columns=TARGET_COLS, errors="ignore")
assert not any(t.lower() in {c.lower() for c in X.columns}
               for t in TARGET_COLS)
```

A unit test ([TODO: tests/test_no_leakage.py: confirm it exists]) runs the same assertion in CI. The same X is used for all four targets, so no target ever sees another target.

Features with ambiguous or secondary leakage risk

[TODO: Review columns such as "Cath"-adjacent or post-diagnosis fields (e.g. any treatment/outcome-like variable) and state explicitly that none remain.]

3. Model Architecture

Four independent **XGBClassifier** models, one per target (Cath/CAD, LAD, LCX, RCA). Tree ensembles were chosen because the data are small, tabular and mixed-type, and because TreeSHAP gives exact, fast attributions.

Configuration

Item	Setting
Class imbalance	Class weighting ([TODO: scale_pos_weight value or sample_weight scheme])
Monotonic constraints	+1 for Age, blood pressure, LDL, FBS, smoking, RWMA; –1 for EF-TTE, HDL. [TODO: list the complete constraint vector and clinical justification for each]
Hyperparameters	[TODO: max_depth, n_estimators, learning_rate, subsample, regularisation; how chosen (inner CV grid?)]
Calibration	[TODO: isotonic / Platt, fitted inside training folds, or state "not applied"]
Baseline	[TODO: logistic regression (L2) with same folds, results in Section 4]
Persistence	[TODO: model files, e.g. models/*.json; seed values]

Why monotonic constraints

Constraints force the model response to move in a clinically plausible direction (higher age raises risk, higher EF lowers it). They improve the sanity of what-if analysis and act as a regulariser on 303 samples. They do **not** make the response causal (see Section 6).

Explainability

shap.TreeExplainer returns per-patient, per-vessel log-odds contributions on every prediction request. The dashboard groups contributions by category (demographics, vitals, laboratory, ECG, echo), shows the top drivers, and marks each feature as modifiable or non-modifiable. SHAP values describe *how the model reached its output*, not biological causation.

Probability-to-colour mapping

Vessel colour: green below 25%, amber 25–49%, red at 50% or above. These cut-offs are display conventions. [TODO: state whether probabilities are calibrated; if not, say the 50% colour threshold is a visual convention and not a calibrated risk statement]

4. Evaluation

Protocol. [TODO: describe exactly: stratified 5-fold CV (repeats?), random seed, whether hyperparameters were tuned on the same folds used for reporting. If yes, say so; if nested CV was added, describe inner/outer loops]. All metrics are computed on out-of-fold predictions only. Threshold for precision/recall/F1: [TODO: 0.5 or tuned? how?].

Results (out-of-fold)

Target	ROC-AUC (95% CI)	Accuracy	Precision	Recall	F1	Brier
Overall CAD	0.9085 ([TODO: CI])	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
LAD	0.8474 ([TODO: CI])	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
LCX	[TODO: AUC (CI)]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RCA	[TODO: AUC (CI)]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Baseline LR, CAD	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Only the CAD (0.9085) and LAD (0.8474) ROC-AUC values come from the team's completed runs; every other cell must be filled from your own output before submission. Do not estimate them.

Interpretation

- AUC around 0.9 for CAD is in the range reported for this dataset in the literature; it is not near-perfect, which is consistent with no leakage.
- LAD is the best-represented vessel signal; LCX and RCA are expected to be weaker and noisier because of fewer positive cases. [TODO: report actual numbers and discuss].
- [TODO: Add ROC curves and a reliability (calibration) plot as a figure here; one image for all four targets saves space.]
- [TODO: Add a short SHAP summary figure for CAD showing the top features and check they agree with clinical expectations (e.g. typical chest pain, age, ECG changes).]

5. 3D Visualization Pipeline and Integration

3D scene

Built with Three.js (WebGL) and no build step. The scene contains a heart model and three coronary vessel trees. Vessel paths are [TODO: describe: splines (CatmullRomCurve3 + TubeGeometry) placed on an imported mesh, or on procedural geometry]. Intended routing: LAD in the anterior interventricular groove, LCX in the left atrioventricular groove, RCA in the right atrioventricular groove toward the crux. [TODO: state how routing was verified, e.g. checked against an anatomy reference].

Risk mapping and interaction

- Each vessel material colour is set from its predicted stenosis probability returned by /api/predict (no hard-coded values).
- Orbit, zoom and pan via OrbitControls; Raycaster click-picking selects a vessel and opens its detail panel (probability, top SHAP drivers).
- Preset camera views (AP, LAO, RAO) for familiar angiographic orientation. These are viewing angles only and do not simulate fluoroscopy.
- Stenosis markers and flow particles are visual aids driven by the probability. They indicate model output, not a located lesion.

What the 3D view does not claim

The dataset provides no lesion coordinates or imaging. The model predicts *whether* a vessel has ≥50% stenosis, not *where* along it. The marker position on a vessel is a fixed illustrative location. Region with RWMA is used only as a model feature and is not mapped onto the anatomy.

Data flow

Browser form (or parsed report) → POST /api/predict → FastAPI validates with Pydantic → four XGBoost models + TreeSHAP → JSON (CAD probability, vessel probabilities, SHAP per vessel) → dashboard and 3D scene update. /api/init supplies feature metadata and defaults so the form is generated from configuration.

Performance

Inference latency: [TODO: measure and report median and p95 over N requests, e.g. with a timing script; remove the "sub-20 ms" claim if not measured]. Rendering: [TODO: measured FPS on a laptop without a dedicated GPU, plus browser and resolution].

Extensibility

Feature lists, constraints and vessel definitions live in configuration. Adding a feature means updating the schema and retraining; adding a model means registering one more target; adding a vessel means adding a curve definition and a mapping entry.

6. Usage, Limitations and Safety

Setup and usage

```
git clone <repo-url>
cd CardioTriage3D
pip install -r requirements.txt
python backend/inspect_and_train.py     # trains models, writes metrics
uvicorn backend.main:app --reload       # serves API and frontend
# open http://localhost:8000
```

[TODO: Confirm commands, file names and port against the actual repo before submitting.] Workflow: enter or load patient features → Predict → inspect CAD status and vessel probabilities → rotate/zoom/click vessels → review SHAP breakdown → adjust modifiable features in the what-if panel.

What-if sensitivity analysis

The panel lets a user change modifiable inputs (BP, LDL, FBS, BMI, smoking) and see how the model output changes. This shows the **model's learned association**, not a treatment effect: the data are cross-sectional, no patient was randomised, and existing stenosis does not regress because a lab value changes. Results must not be read as expected benefit of therapy.

Limitations

- **Small, single-centre dataset** (n = 303); no external validation, so performance on other populations is unknown.
- **Few positives for some vessels**, giving wide uncertainty for LCX and RCA.
- **Cross-sectional labels** from a single angiographic assessment; no outcomes or follow-up.
- **Binary 50% threshold labels**; no stenosis severity or lesion location.
- **Vessel labels are correlated** but modelled independently; displayed vessel results may not be mutually consistent. [TODO: describe the consistency check, if any, in plain terms].
- **Calibration:** [TODO: state status].
- **Tuning bias:** [TODO: state whether reported metrics could be optimistic due to tuning on the same folds].

Clinical safety statement

CardioTriage 3D is a research and educational prototype for decision-support demonstration. It is not approved for clinical use, must not be used to rule out or diagnose coronary disease, and does not replace coronary angiography, CT angiography, or clinician judgement. A persistent banner in the interface states this on every screen.

References and credits

Dataset: Alizadehsani, R. et al., Z-Alizadeh Sani / Extension, UCI ML Repository (ID 411). [TODO: add exact citation]. Libraries: XGBoost, SHAP, scikit-learn, FastAPI, Three.js. 3D assets: [TODO: mesh name, author, license].